Funkcje analityczne

Baza danych: MS SQL

Przemysław Starosta – 2026

Rollup

	department_id	salary
1	NULL	7000.00
2	10	4400.00
3	20	19000.00
4	30	24900.00
5	40	6500.00
6	50	156400.00
7	60	28800.00
8	70	10000.00
9	80	304500.00
10	90	58000.00
11	100	51600.00
12	110	20300.00
13	NULL	691400.00

```
select
    department_id,
    sum(salary)
    from employees
group by
rollup(department_id)
```

Rollup – wiele grup

	department_id	job_id	sum
1	10	AD_ASST	4400.00
2	10	NULL	4400.00
3	20	MK_MAN	13000.00
4	20	MK_REP	6000.00
5	20	NULL	19000.00
6	30	PU_CLERK	13900.00
7	30	PU_MAN	11000.00
8	30	NULL	24900.00
9	NULL	NULL	48300.00

```
select
    department_id,
    job_id,
    sum(salary) sum
    from employees
where department_id in (10,20,30)
group by rollup(department_id, job_id)
```

Cube

	department_id	sum
1	10	4400.00
2	20	19000.00
3	30	24900.00
4	NULL	48300.00

```
select
    department_id,
    sum(salary) sum
    from employees
where department_id in (10,20,30)
group by cube(department_id)
```

Cube – wiele grup

	department_id	job_id	sum
1	10	AD_ASST	4400.00
2	NULL	AD_ASST	4400.00
3	20	MK_MAN	13000.00
4	NULL	MK_MAN	13000.00
5	20	MK_REP	6000.00
6	NULL	MK_REP	6000.00
7	30	PU_CLERK	13900.00
8	NULL	PU_CLERK	13900.00
9	30	PU_MAN	11000.00
10	NULL	PU_MAN	11000.00
11	NULL	NULL	48300.00
12	10	NULL	4400.00
13	20	NULL	19000.00
14	30	NULL	24900.00

```
select
    department_id,
    job_id,
    sum(salary) sum
    from employees
where department_id in (10,20,30)
group by cube(department_id, job_id)
```

Grouping

	department_id	sum	grouping
1	10	4400.00	0
2	20	19000.00	0
3	30	24900.00	0
4	NULL	48300.00	1

```
select
    department_id,
    sum(salary) sum,
    grouping(department_id) grouping
    from employees
where department_id in (10,20,30)
group by rollup(department_id)
```

Grouping – wiele grup

department_id	job_id	sum	grouping_dept	grouping_job
10	AD_ASST	4400.00	0	0
10	NULL	4400.00	0	1
20	MK_MAN	13000.00	0	0
20	MK_REP	6000.00	0	0
20	NULL	19000.00	0	1
30	PU_CLERK	13900.00	0	0
30	PU_MAN	11000.00	0	0
30	NULL	24900.00	0	1
NULL	NULL	48300.00	1	1

```
select
    department_id,
    job_id,
    sum(salary) sum,
    grouping(department_id) grouping_dept,
    grouping(job_id) grouping_job
    from employees
where department_id in (10,20,30)
group by rollup(department_id, job_id)
```

Grouping – null interpretation

	department_id	job_id	sum	grouping_dept	grouping_job
1	10	AD_ASST	4400.00	0	0
2	10	all job_id	4400.00	0	1
3	20	MK_MAN	13000.00	0	0
4	20	MK_REP	6000.00	0	0
5	20	all job_id	19000.00	0	1
6	30	PU_CLERK	13900.00	0	0
7	30	PU_MAN	11000.00	0	0
8	30	all job_id	24900.00	0	1
9	all department	all job_id	48300.00	1	1

```
select
    case when GROUPING(department_id)=1 then 'all department'
else CONVERT(varchar, department_id) end as department_id,
    case when GROUPING(job_id)=1 then 'all job_id'
        else CONVERT(varchar, job_id) end as job_id,
    sum(salary) sum,
    grouping(department_id) grouping_dept,
    grouping(job_id) grouping_job
    from employees
where department_id in (10,20,30)
group by rollup(department_id,job_id)
```

Grouping_id

	department_id	job_id	sum	grouping
1	10	AD_ASST	4400.00	0
2	NULL	AD_ASST	4400.00	2
3	20	MK_MAN	13000.00	0
4	NULL	MK_MAN	13000.00	2
5	20	MK_REP	6000.00	0
6	NULL	MK_REP	6000.00	2
7	30	PU_CLERK	13900.00	0
8	NULL	PU_CLERK	13900.00	2
9	30	PU_MAN	11000.00	0
10	NULL	PU_MAN	11000.00	2
11	NULL	NULL	48300.00	3
12	10	NULL	4400.00	1
13	20	NULL	19000.00	1
14	30	NULL	24900.00	1

```
select
    department_id,
    job_id,
    sum(salary) sum,
    grouping_id(department_id,job_id) grouping
    from employees
where department_id in (10,20,30)
group by cube(department_id,job_id)
```

Grouping sets

	department_id	job_id	sum
1	10	AD_ASST	4400.00
2	NULL	AD_ASST	4400.00
3	20	MK_MAN	13000.00
4	NULL	MK_MAN	13000.00
5	20	MK_REP	6000.00
6	NULL	MK_REP	6000.00
7	30	PU_CLERK	13900.00
8	NULL	PU_CLERK	13900.00
9	30	PU_MAN	11000.00
10	NULL	PU_MAN	11000.00
11	10	NULL	4400.00
12	20	NULL	19000.00
13	30	NULL	24900.00

```
select
    department_id,
    job_id,
    sum(salary) sum
    from employees
where department_id in (10,20,30)
group by grouping sets((department_id), (department_id, job_id),
(job_id))
```